Spectroscopic analysis of Quantum confinement and size distribution in 0D materials

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Abstract

This report is a complete optical characterization of CdSe core and ZnS shell QDs, paying attention to the quantitative analysis of the quantum confinement effect. Spectral data were extracted from the provided Group K data in the range of 400-800 nm wavelength, and the background noise was subtracted from the absorbance data, through which the excitonic peaks were extracted. The analysis revealed the major excitonic peaks at 523.1 nm and 593.2 nm, which are associated with the E_g of 2.37 eV (Green) and 2.09 eV (Orange). By applying the empirical model, these optical signatures were converted to real-life QD diameters of 2.47 ± 0.52 nm and 4.14 ± 0.65 nm.

In addition to the experimental characterization, the synthesis landscape and the use of QDs in the sophisticated management of photons are also assessed in this work. Although the traditional hot injection is the norm in monodispersity, recent trends focus on AI-driven and heavy-metal-free compositions to improve sustainability. The report also describes the working principle of LDS, illustrating how the QDs enhance the EQE of PV solar cells by aligning the incident UV/blue photons with the maximum spectral response of silicon. Lastly, the importance of ZnS shell in offering surface passivation is also discussed, and it is found to be essential in avoiding non-radiative recombination. This confirms the effectiveness of CdSe/ZnS QDs as tunable materials in high-efficiency optoelectronic and solar energy harvesting systems.

I. INTRODUCTION

Quantum dots (QDs) are an important breakthrough in nanotechnology, as they are zero-dimensional (0D) nanoparticles in which charge carriers, i.e. electrons and holes, are confined in all three spatial dimensions. This report focuses on the optical characterization of Cadmium Selenide (CdSe) core and Zinc Sulfide (ZnS) shell QDs [2]. In contrast to bulk semiconductors, in which the energy bands are continuous, the extreme confinement of QDs results in discrete atom-like energy levels [1].

Table 1 Quantum dot synthesis trends

Methods	Advantages	Disadvantages
Hot injection	Superior monodispersity and crystallinity	Requires hazardous reagents and high temperature.
Aqueous synthesis	Eco-friendly and biocompatible	Generally, lower quantum yields than organic routes.
AI-driven synthesis	Uses machine learning to predict optimal crystal growth.	Requires massive initial datasets for training.

Recent advancements have seen the rise of heavy-metal-free QDs, such as InP or Carbon QDs, to replace 'lead and cadmium' based materials [4].

Background

Quantum confinement is a phenomenon that happens when the physical size of a semiconductor crystal is reduced to less than the Bohr exciton radius, and the system's energy levels change. The smaller the diameter of the QD, the larger the energy gap (E_g) between the conduction band and the valence band. The colour of the material can be tuned by the shift, with the dots of smaller size emitting and absorbing shorter (blue) wavelengths, and larger dots shifting to longer (red) wavelengths [1][3].

The samples used in this lab consist of a CdSe core encapsulated by a ZnS shell. The ZnS shell has a wider bandgap, which passivates surface states and eliminates dangling bonds that would otherwise act as non-radiative recombination sites. Such a structure significantly improves the photoluminescence quantum yield and chemical stability of the dots.

- **Downshifting mechanism:**

Luminescent Downshifting (LDS) is a photon management process that improves the spectral match between a light source and a receiver, such as a solar cell. In this process, the QD absorbs high-energy photons, which excite electrons into higher energy states. Following this, the electrons undergo non-radiative relaxation, losing a small portion of energy as heat through phonon emission, as they settle at the conduction band edge. Finally, the system achieves emission by releasing a lower-energy photon, which is more efficiently converted by the underlying device.

Critical features impacting LDS efficiency include a large Stokes shift to minimize self-absorption and high surface passivation to prevent non-radiative recombination at surface defects [5][2].

II. METHODOLOGY

A. *Optical setup and components*

The characterization was performed using a customized spectroscopy bench.

- **Light source:** A halogen lamp providing a broadband spectrum from 300 nm to 1100 nm.
- **Fiber optics:** Light is transmitted through modern optical fibers because these are preferred for higher efficiency and fewer core inconsistencies.
- **Collimation lenses:** Two lenses are used to parallelize the light beam before it interacts with the sample.
- **Spectrometer:** This is a device which has an internal grating that disperses light into its individual wavelengths.

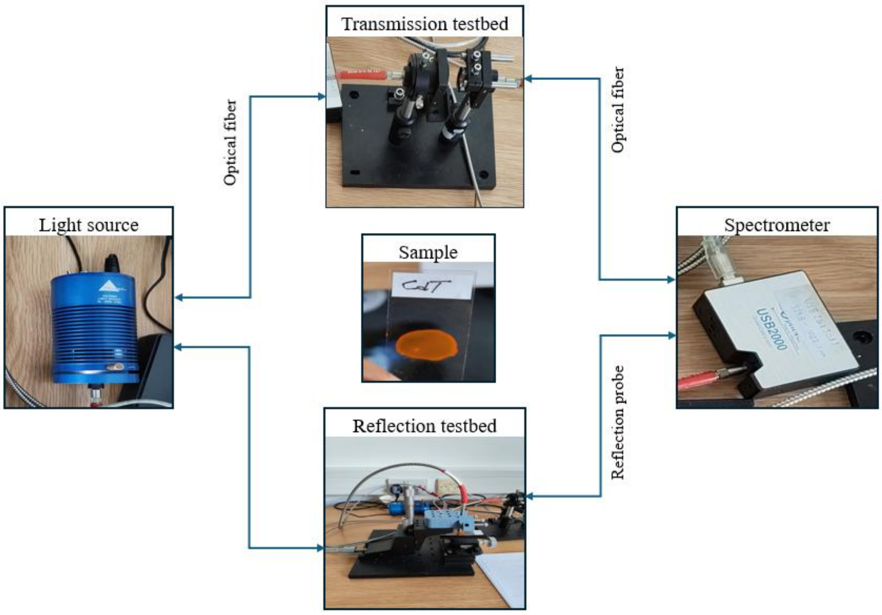


Figure 1 Representation of the measurement setup, which is done in the lab

B. *Software and tools*

The main software that was used in the data acquisition was OceanView, which controlled some important parameters such as:

- **Integration time:** The sensor collects light over the course of this time. Adjustments in this may increase intensity but also can cause saturation.
- **Scan-to-average:** The number of discrete measurements averaged together to reduce signal noise and uncertainty.
- **Boxcar width:** This is a smoothening technique that averages adjacent pixels. While it reduces noise, excessive use can blur physical spectral features.
- **Electric dark:** A mode to remove the dark current, the electric noise that is produced by the heat of the light source and the surroundings.
- **Counts:** The photon tally on a pixel, which shows a linear relationship with integration time unless saturated.
- **Non-linearity correction:** Addresses instances where counts do not maintain a linear relationship with intensity across certain wavelengths, necessitating a corrective function.

C. *Data acquisition procedure*

The experiment was separated into three different phases of measurements in order to isolate the optical characteristics of the QDs. To obtain absolute values, baseline measurements were done:

- **Dark spectrum:** Measured with the light source turned off to determine electronic background noise.
- **Transmittance (T):** To depict 100% transmission, all measurements are made using air or a blank substrate. The QD sample was put between the collimation lenses. In this step, it was observed that the focal point can be moved by changing the refractive index of the glass substrate, leading to an artificial increase in the measured intensity. This must be corrected by normalizing the integration time or recalibrating the focus.
- **Reflectance (R):** A gold sheet or silicon wafer was placed to offer a known, high-reflectivity reference. This parameter was measured by directing light onto the sample surface and collecting the back-scattered light.

$$R = \frac{Sample\ count}{Silicon\ reference\ count} \quad (i)$$

D. Calculations

Based on the provided data for Group K Green and Orange QDs, we analyze the Transmission and Reflection in the reference range of 400 nm to 800 nm using MATLAB:

- Transmittance data: It shows a characteristic drop in transmission at wavelengths below 500 nm, indicating the onset of electronic transitions within the QDs.
- Reflectance data: The reflection remains relatively low and stable, characteristic of thin-film QD layers on glass.

Background noise calibration:

Before sample measurement, the system's 'dark current' was quantified with the light source deactivated and the average intensity value was taken across the entire wavelength spectrum. Because the noise can accumulate over time, the background noise must be considered in relation to the integration time set in the software.
This value was subtracted from the raw intensities to isolate the true optical response of the QDs.

Absorbance calculation:

To find the absorbance (A), we use the principle of conservation of energy. The formula for "A" used in this lab is

$$A = 1 - T - R \quad (ii)$$

In this lab report, we prioritize the total absorbed energy fraction to account for scattering and reflection losses. To find the peak of QDs, we performed a polynomial fit, which smooths out the noise spikes while preserving the actual shape of the excitonic feature.

Bandgap (E_g) calculation:

To find the E_g, we used a Tauc plot, which is based on the relationship between the absorption coefficient and the energy of the incident photon. The core principle is that near the absorption edge, the absorption coefficient follows a power-law relationship.

$$(\alpha h\nu)^{1/n} = A\left(h\nu - E_g\right) \quad (iii)$$

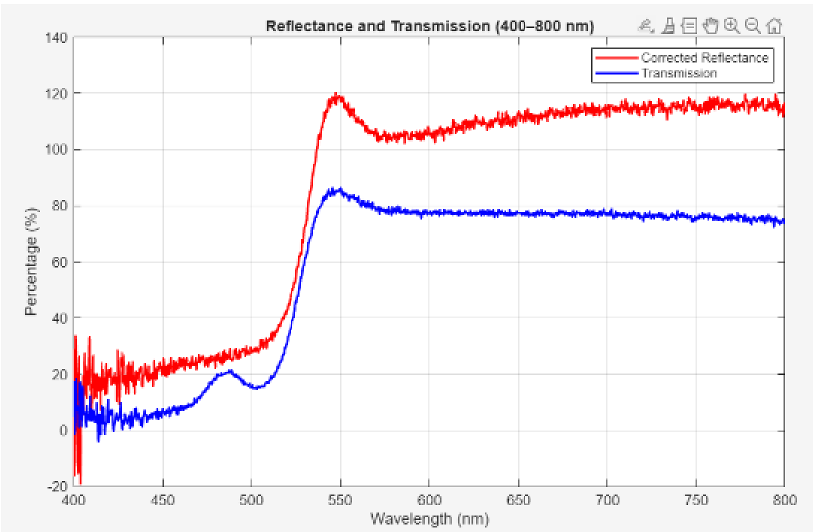
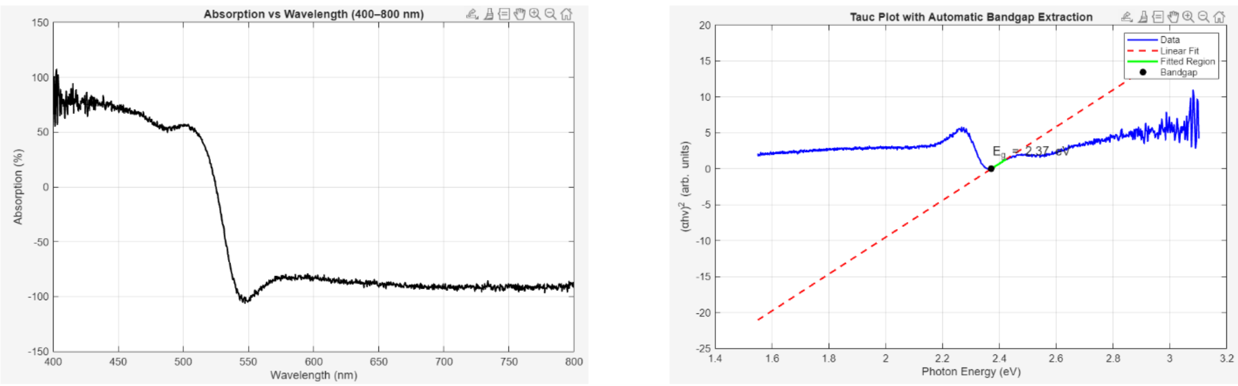


Figure 2 Corrected Green QD reflectance with respect to the reflectivity of Si



(a) (b)

Figure 3 (a) Absorbance of Green QD (b) Extracted E_g = 2.37 eV from the Tauc plot

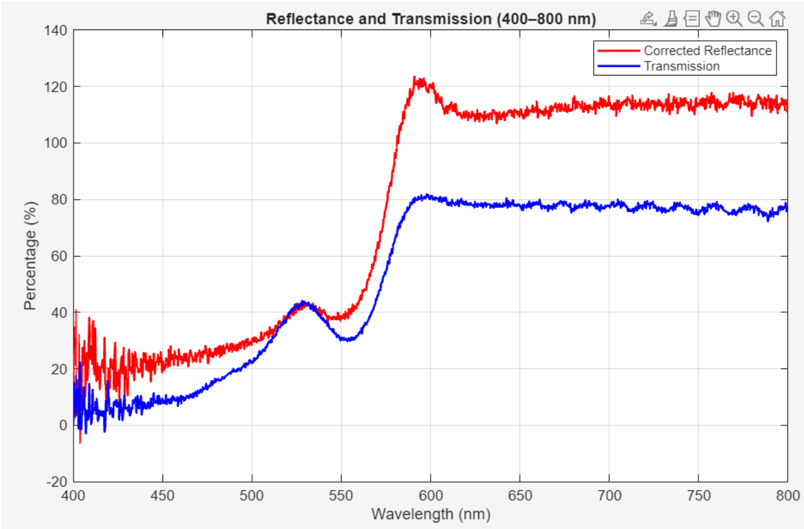
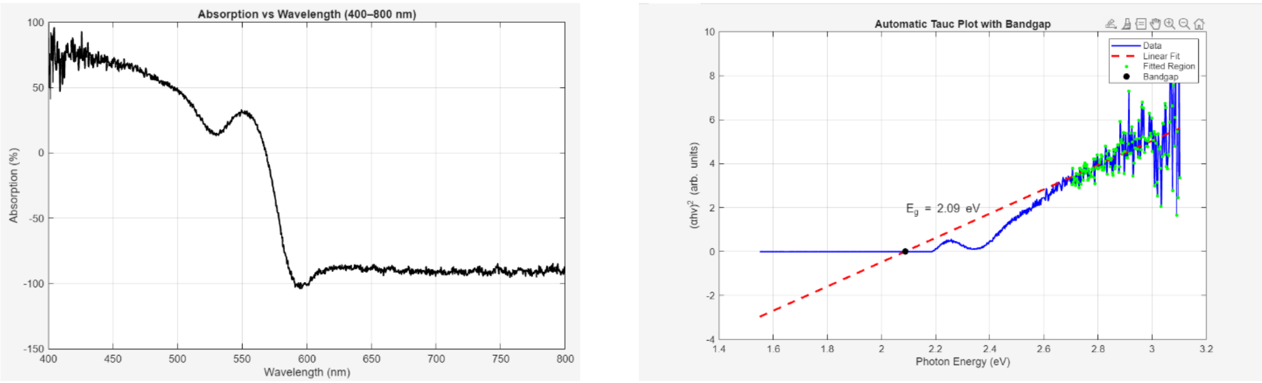


Figure 4 Corrected Green QD reflectance with respect to the reflectivity of Si



(a) (b)

Figure 5 (a) Absorbance of Orange QD (b) Extracted E_g = 2.09 eV from the Tauc plot

The peak is identified in the absorbance spectrum, which was derived at every wavelength using formula (iii). By observing the data, it was found that the absorbance reached its maximum value at:

- Green QDs (peak ~523.1 nm) Orange QDs (peak ~593.2 nm):

This confirmed that 523.1 nm and 593.2 nm are the wavelengths where the Green QD and Orange QD have their maximum interaction with incident photons, representing the discrete energy level of their bandgap.

The higher absorbance in Green QDs indicates a stronger oscillator strength at shorter wavelengths, consistent with their smaller physical diameter and larger bandgap.

The E_g can also be calculated using the first excitonic absorption peak, which corresponds to the lowest energy transition.

$$E_g(eV) = \frac{hc}{\lambda_{peak}} \quad (iv)$$

QD size calculation:
To determine the diameter of the QDs, we look at the relationship between quantum confinement and particle size.
The empirical formula, which is mentioned in reference [6]

$$D = (1.6122 * 10^{-9})\lambda^4 - (2.6575 * 10^{-6})\lambda^3 + (1.6242 * 10^{-3})\lambda^2 - (0.4277)\lambda + 41.57 \quad (v)$$

Similarly, the size distribution (σ_D) is calculated from the absorbance spectrum by measuring the Full Width at half Maximum (FWHM) of the excitonic peaks, and calculating the Spectral Standard Deviation (σ_λ) [15]:

$$\sigma_D = \left|\frac{dD}{d\lambda}\right| \sigma_\lambda \quad (vi)$$

Green QDs: Absorption peak occurs at 523.1 nm, yielding a diameter of 2.47 nm and $\sigma_D = \pm 0.52\ nm$, which corresponds to an E_g of 2.37 eV.
Orange QDs: Absorption peak occurs at 593.2 nm, yielding a diameter of 4.14 nm and $\sigma_D = \pm 0.65\ nm$, which corresponds to an E_g of 2.09 eV.

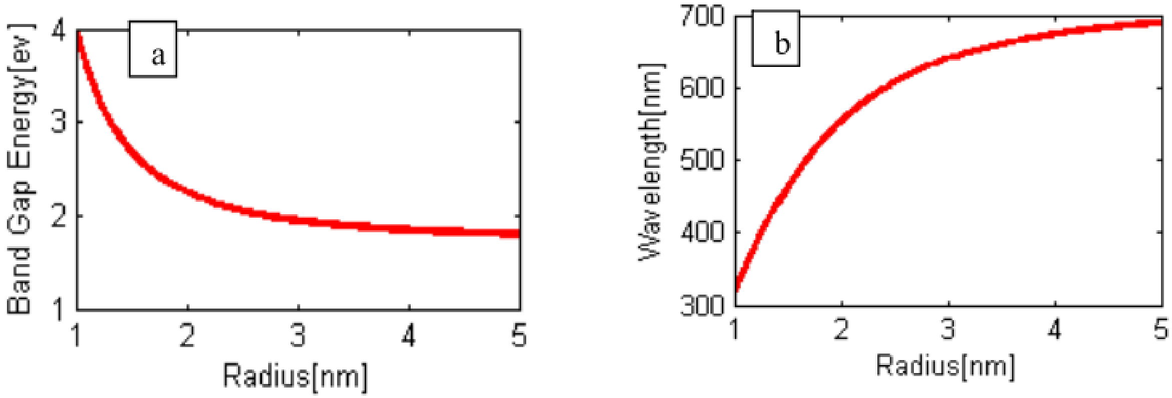


Figure 6 (a). Different sizes of CdSe quantum dots vs. Bandgap energy (b). Different sizes of CdSe quantum dots with their corresponding Emission Wavelength [8]

E. Error analysis

The intensity spike which happens upon the insertion of the sample is crucial. The glass substrate is a refractive medium that alters the optical path and concentrates more light in the collection fiber than when using air alone as the medium, causing an artificial jump in intensity that is not related to QDs themselves. The absorption may show negative values because of the "refractive glass effect", in which the glass side focuses more light into the fiber than the initial air reference.

To achieve a physical response, the integration time has to be reduced to restore the signal to the linear region of the spectrometer.

- The halogen lamp is hot. This heat creates "dark current" in the sensors, which must be subtracted to avoid fake data.
- The older multi-mode fibers used in the lab have larger cores that are less efficient than modern, precision-aligned fibers.
- The thermal energy within the charged pixels generates a baseline signal even at complete darkness. Failure to subtract this would lead to an underestimation of absorbance.
- Reflections from the lab bench or environmental lighting were mitigated by using a black sheet over the setup.

III. RESULTS

The colour of the QD is determined by its "photoluminescence". Using the relationship between E_g, λ, and colour, it is observed that the results found are consistent with theoretical literature and that the smaller (Green) dot has a higher E_g due to stronger confinement and requires more energetic photons to trigger an electronic transition [6].

A. Setup improvements

To improve the accuracy of the setup, we must address challenges related to spatial resolution, SNR, and the optical movement of the sample:

- Utilizing a high NA objective lens to focus the probing beam to a diffraction-limited Gaussian spot size, which allows for precise alignment over a single QD and provides a well-defined intensity profile [9]. Using a solid immersion lens (SIL) reduces the spot area, which leads to improved SNR and signal resolution [11].
- Incorporating a Distributed Bragg Reflector (DBR) and an epitaxially grown gate can lead to reduced background light and improved contrast [10].
- Using a laser would allow for more precise measurement of photoluminescence and emission peaks [10].
- Use one optimized path for transmission and a separate, high-angle path for reflection to avoid refractive artefacts.

B. Enhancing EQE via LDS

Most commercial solar cells exhibit poor EQE at short wavelengths, i.e. 300-450 nm, due to high surface recombination and parasitic absorption by front and side layers like glass [7].

The mechanism to enhance this is that a QD-doped layer, often integrated into the encapsulation, absorbs these UV photons and re-emits them as visible light. This visible light travels into the cell, where the silicon has a much higher response, effectively shifting unusable energy into a usable range. LDS shifts the energy much closer to the bandgap edge of silicon, which reduces the thermalization loss where excess photon energy is wasted as heat through phonon emission.

C. Beneficiary solar cell technologies

1. **C-Si solar cells:** LDS shifts the blue-light photons to the visible range, bypassing the poor response and high front-surface recombination.
2. **Thin-film (CdTe, CIGS) solar cells:** These materials have a wide bandgap that shades the active material from UV light. LDS can bypass this by shifting UV light into the active absorption band.
3. **Perovskite solar cell:** LDS provides UV stability, as the QD layer acts as a sacrificial UV filter while still providing useful visible photons to the perovskite layer [4].

D. Mechanisms besides LDS

1. **Multiple Exciton Generation (MEG):** It allows a single high-energy photon to generate two or more e^--h^+ pairs, whereas standard silicon cells only generate one. This occurs because the confinement in QDs reduces the rate of cooling, giving the excited electron enough time to collide with another electron and excite it as well [12].
2. **Intermediate Band Solar Cells (IBSC):** QDs can be used to create a “third band” within the semiconductor’s energy gap. This allows the cell to absorb low-energy infrared photons that would typically pass right through a standard cell, significantly increasing the short-circuit current [13].
3. **Luminescent Solar Concentrators (LSC):** In this technology, the QDs are embedded in a plastic or glass pane. They absorb light from the entire surface area and re-emit it toward the edge, where small, high-efficiency solar cells are located, concentrating the light and reducing the total amount of expensive PV material needed [14].

E. Impact on surface passivation and optics

QDs influence PV performance not just through light conversion, but through material interface quality.

As noted in the background, the encapsulated ZnS shell passivates the dangling bonds at the surface of the QD core. In a PV cell, this prevents non-radiative recombination at the interface, ensuring high yield [2].

QD layers can be engineered to have a refractive index that provides a graded transition between air and the solar cell, effectively acting as an anti-reflective coating that minimizes reflection (R) losses.

IV. CONCLUSION

The experimental data highlight the essential nature of size-dependent bandgap engineering as the determinant of the optical behaviour of QDs. It establishes that through effective isolation of physical absorption of raw spectral data, the ability to predictively manipulate energy levels based on the quantum confinement theory increases. Furthermore, a protective wide-bandgap shell synthesis was essential in neutralizing surface defects. These observations confirm the adoption of these QDs as special spectral converters in PV devices, are useful in reducing the energy losses and surpassing the Shockley-Queisser limit. Finally, the flexibility of these QDs with powerful chemical and optical schemes makes them a base technology for the next generation of high-performance and sustainable optoelectronic devices.

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